# A continental-scale analysis of the genomic and ecological basis of local adaptation in a songbird

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# Abstract

# Introduction

The vast majority of organisms inhabit spatially heterogeneous ecological environments. This environmental variation exerts variable selection pressures, and as a result many species exhibit ecologically induced phenotypic and genetic differentiation, or 'local adaptation'. If reproductive barriers arise between populations facing divergent ecological selection, local adaptation can be a precursor to speciation. Furthermore, understanding local adaptation is necessary for predicting how populations will respond to environmental change. Thus, there is great interest in identifying the extent to which local adaptation occurs in natural populations, the forces that drive it, and the phenotypic and genetic changes involved.

Over the past decade, genomic data have become an invaluable tool for studying local adaptation in wild populations. Studies using genomic data have tended to take one of three approaches. The most commonly employed approach has been to examine patterns of genomic differentiation among populations and identify highly differentiated regions, which are assumed to be subject to divergent selection. If the function of genes in regions involved in local adaptation can be identified, it is sometimes possible to make inferences about the traits involved in local adaptation. Alternatively, if a specific phenotypic trait is thought to be subject to natural selection, genomic approaches can be used to identify the genetic basis of this trait, and patterns of variation at candidate regions can be compared across populations in order to identify whether local adaptation is operating. Finally, patterns of genomic variation can be correlated with one or a range of environmental variables that are candidates for drivers of local adaptation. The advantage of this approach is that it makes it possible to link genotype, phenotype and the driver of selection.

The great tit (*Parus major*) is a widespread passerine found throughout Europe, as well as parts of the Middle East, Asia and North Africa. In Europe, the great tit is thought to have colonised from a single glacial refugium in the Balkans during the last glacial period. The recent colonisation of this species across Europe, combined with very large population sizes and high levels of gene flow, have resulted in high levels of genetic variation and low levels of structure at neutral markers, even at the continental scale.

The great tit genome has recently been sequenced (Laine *et al.* 2016), and a high density SNP chip developed.

Excellent ecological data available for this species, making it a highly tractable model species for studying local adaptation.

In this study we analyse genomic variation across XX great tit populations in Europe. We first quantify levels of genetic variation within and across populations and use this to make inferences about population

demography and colonisation history. We then analyse genomic divergence among populations in order to identify regions of the genome involved in recent adaptation to different environments. Finally, we perform genotype-environment association analyses to identify regions of the genome involved with adaptation to specific environmental gradients, and to identify regions of the genome associated with contental-wide variation in life-history traits.

# Methods

## Sampling and molecular methods

Samples were collected from 28 populations from 21 countries across Europe (Fig. 1A; Table 1). Birds were trapped from nest boxes, or using mist nets, and ringed with a uniquely numbered aliminium ring. Blood was taken via brachial venipuncture, and stored in either 1 ml Cell Lysis Solution (Gentra Puregene Kit, Qiagen, USA) or Queen's buffer. All samples were genotyped using a custom-made Affymetrix® great tit 650K SNP chip at Edinburgh Genomics (Edinburgh, United Kingdom). After filtering, a total of 647 samples typed at 483888 SNPs were retained for analysis.

## Analysis

Unless stated otherwise, all analyses were carried out in R version 3.3 (R Development Core Team 2011). For population genetic analyses (i.e. all tests not involving single-marker or windowed statistics), we used a pruned and thinned dataset, created in PLINK version 1.9 (Purcell *et al.* 2007). The data were pruned based on linkage disequilibrium to remove all markers with $R^2 > 0.1$, then thinned with a probability of retaining each variant of 0.25. We calculated observed and expected heterozygosity for each SNP and population in the pruned dataset, and calculated $F_{ST}$ between each pair of populations, using PLINK (and custom python scripts to loop over populations). Pairwise $F_{ST}$ was compared to geographic distance between populations using Mantel tests, implemented in the Ecodist package in R (Goslee & Urban 2007). We also estimated population structure using Admixture version 1.3, with default settings (Alexander, Novembre & Lange 2009). We varied values of $K$ from two to ten; by which time increasing values of $K$ provided no informative information about population structure (see results). Model support for each value of $K$ was estimated by calculating 5-fold cross-validation error. Finally, we visualised the evolutionary history among European great populations by generating a maximum likelihood tree in TreeMix version 1.13 (**???**). We fitted models allowing for range of migration events (XX-YY), and used a jackknife approach to determine whether migration events significantly improved model fit (**???**).

Marker-based FST

Recombination

## Results

Levels of genetic diversity, heterozygosity

Levels of structure

## Discussion

## References

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## Figure legends

**Figure 1** Sampling locations, heterozygosity

**Figure 2** Population structure in European Great tit populations. **A** Pairwise $F_{ST}$ between all European populations in relation to pairwise geographic distance; **B** pairwise $F_{ST}$ between a proposed refugial population in Turkey and all other populations, in relation to geographic distance from the refugia. Blue points in **A** and **B** are comparisons involving the island populations of Corsica, Sardinia and Crete, while grey points represent all other comparisons. **C** Output from admixture analysis at $K = 6$.

**Figure 3** Manhattan plots

**Figure 4** Bayenv analyses

**Figure 1**

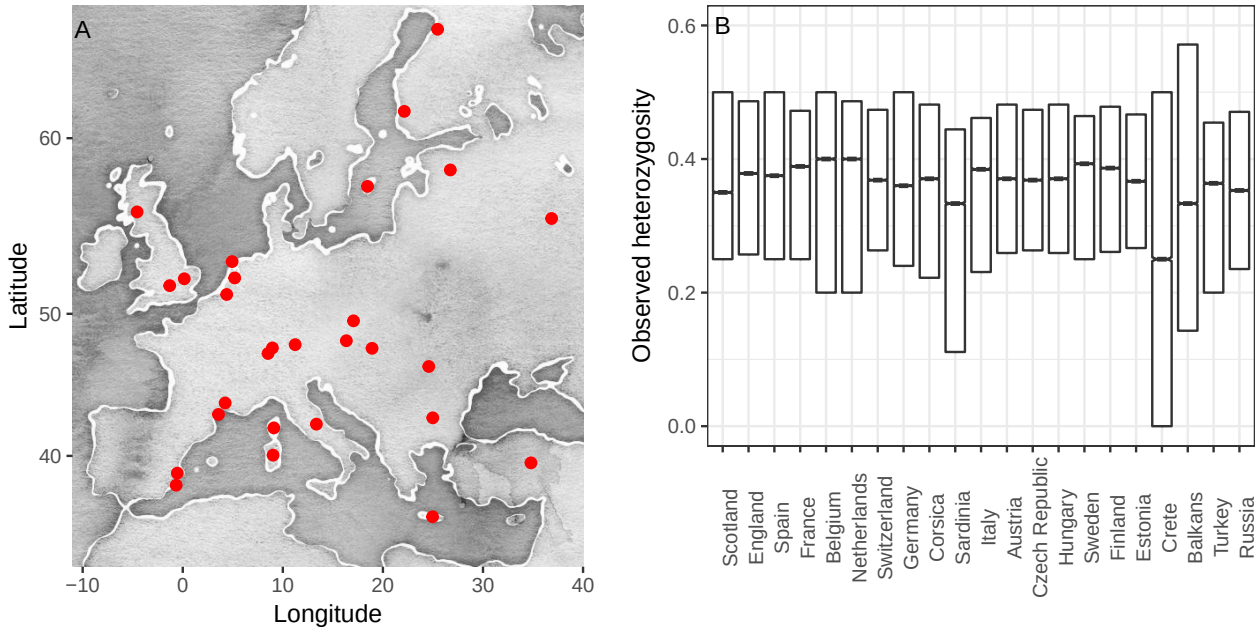
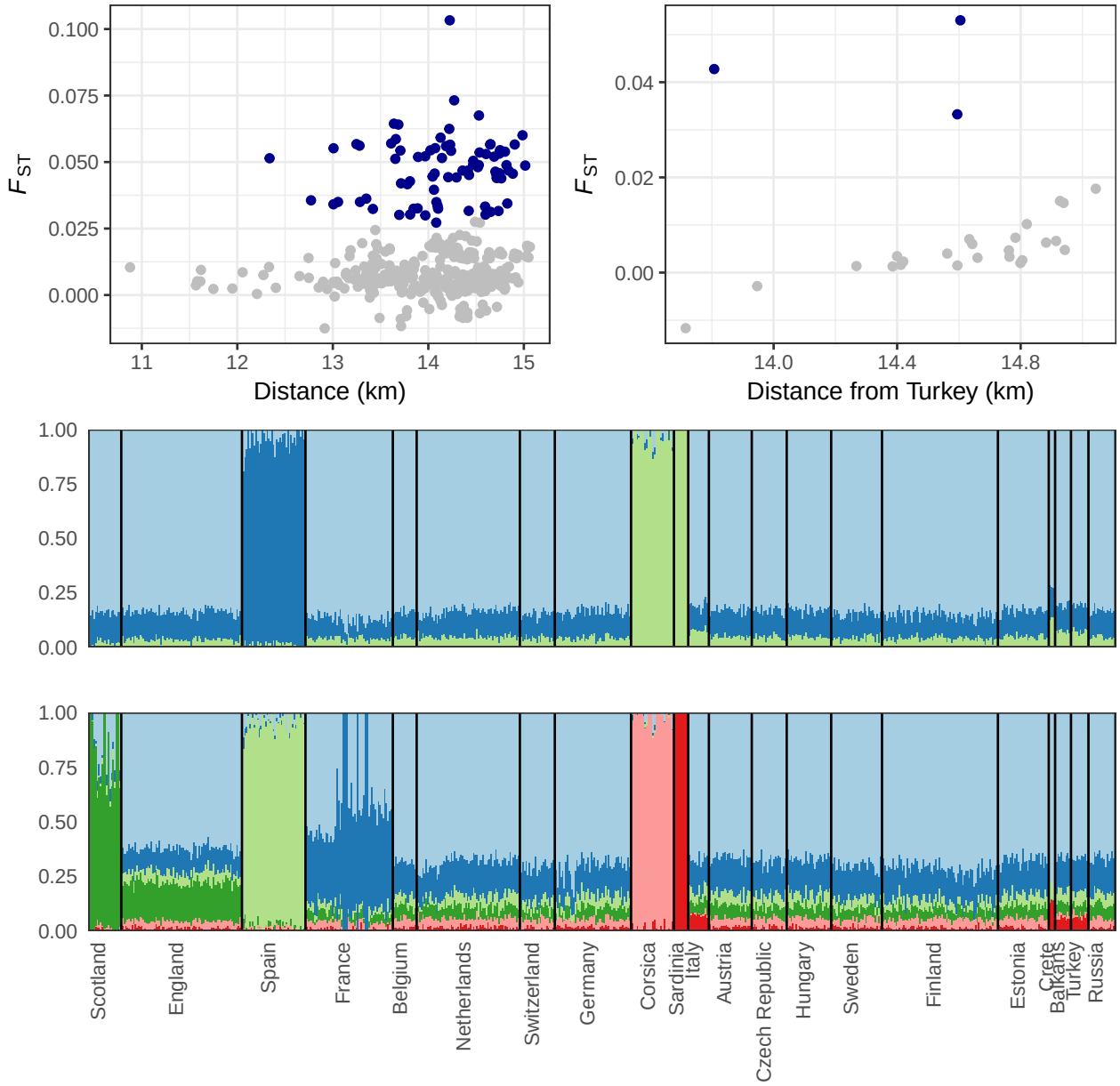

A
Latitude
60
50
40
−10
0
10
20
30
40
Longitude
B
Observed heterozygosity
0.6
0.4
0.2
0.0
Scotland
England
Spain
France
Belgium
Netherlands
Switzerland
Germany
Corsica
Sardinia
Italy
Austria
Czech Republic
Hungary
Sweden
Finland
Estonia
Crete
Balkans
Turkey
Russia

**Figure 2**


0.100
0.075
0.050
0.025
0.000
$F_{ST}$
11
12
13
14
15
Distance (km)
0.04
0.02
0.00
$F_{ST}$
14.0
14.4
14.8
Distance from Turkey (km)
1.00
0.75
0.50
0.25
0.00
1.00
0.75
0.50
0.25
0.00
Scotland
England
Spain
France
Belgium
Netherlands
Switzerland
Germany
Corsica
Sardinia
Italy
Austria
Czech Republic
Hungary
Sweden
Finland
Estonia
Crete
Balkans
Turkey
Russia